

Predictors of Earnings Risk with Machine Learning

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Abstract

This paper looks at the predictors of lifetime earnings risk under a Restricted Income Profile (RIP) model using traditional and machine learning methods such as lasso and SHAP values. The paper builds on the work of Drewianka and Oberg (2025) which uses a moment condition approach derive a parameter that captures permanent income risk. The paper finds that education and age are important in explaining lifetime earnings risk. The paper also finds that macroeconomic variables such as probability of recession and real GDP growth are important and along with state controls may further imply a role of government policy. Finally, the paper finds that occupation controls are important while industry controls do not appear to play a strong role.

Keywords: machine learning, restricted income profile, earnings instability, risk

JEL Codes: D8, J0, D3

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- 1 Introduction
- 2 Theoretical Framework
- 3 Data
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 - 5.1 Permanent Shocks

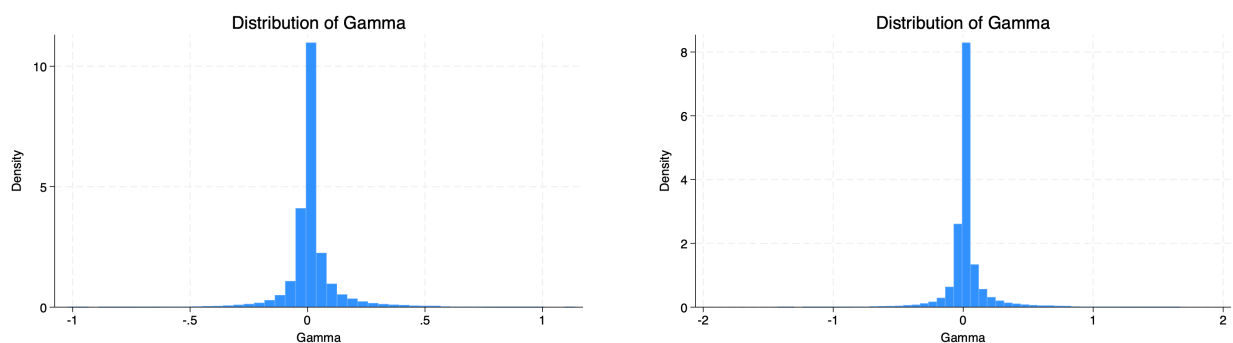


Figure 1: Distribution of Gamma: Hourly earnings (fhwage, left) and Annual earnings (fearn, right)

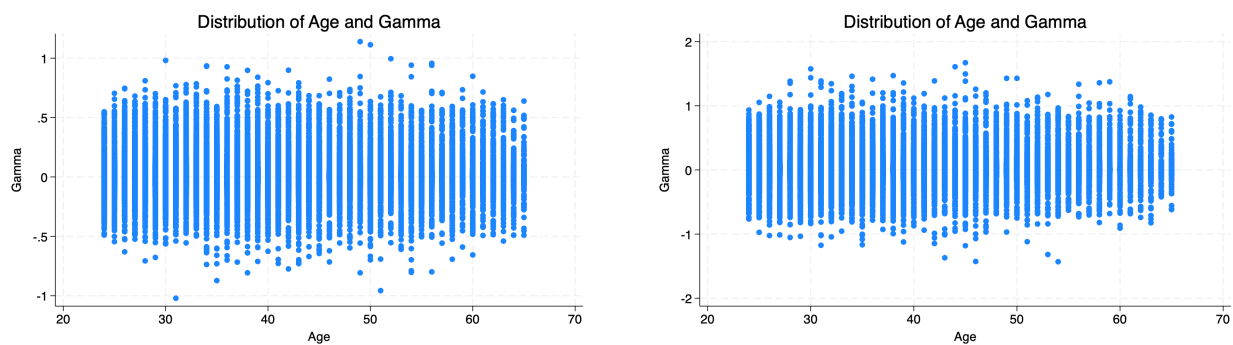


Figure 2: Scatterplot of Age vs. Gamma: Hourly earnings (fhwage, left) and Annual earnings (fearn, right)

Table 1: Gamma Regressions: F-Test Results (Hourly Earnings, fh wage)

	No Occ/Ind (1)	No Occ (2)	No Ind (3)	All Controls (4)
Census Division FE	1.98** (0.0371)	1.95** (0.0406)	1.81* (0.0605)	1.71* (0.0816)
Year FE	1.19 (0.2125)	1.16 (0.2411)	1.18 (0.2249)	1.14 (0.2680)
Race FE	1.59 (0.1582)	1.18 (0.3150)	1.09 (0.3645)	1.13 (0.3406)
Cohort FE	3.92*** (0.0083)	3.85*** (0.0091)	4.19*** (0.0057)	4.13*** (0.0061)
Occupation FE	—	—	2.59*** (0.0000)	2.20*** (0.0000)
Industry FE	—	2.93*** (0.0000)	—	2.01*** (0.0006)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

Table 2: Gamma Regressions: F-Test Results (Annual Earnings, fearn)

	No Occ/Ind (1)	No Occ (2)	No Ind (3)	All Controls (4)
Census Division FE	2.98*** (0.0015)	2.77*** (0.0031)	2.92*** (0.0018)	2.58*** (0.0057)
Year FE	1.48** (0.0384)	1.82*** (0.0030)	1.73*** (0.0064)	1.79*** (0.0040)
Race FE	0.71 (0.6174)	0.55 (0.7350)	0.64 (0.6703)	0.71 (0.6170)
Cohort FE	2.01 (0.1106)	2.37* (0.0682)	2.16* (0.0901)	2.33* (0.0718)
Occupation FE	—	—	2.50*** (0.0000)	1.88*** (0.0000)
Industry FE	—	3.69*** (0.0000)	—	2.19*** (0.0001)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

Table 3: Gamma Regressions: Industry Selection Results

Hourly Earnings (flwage)				Annual Earnings (fearn)			
Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order	Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Public Administration	5	1	9	Not Applicable	1	1	1
Not Applicable	1	2	2	Constr. Relate	6	2	3
Construction	4	3	1	Public Administration	4	3	24
Clothing/Text	18	4	27	Other Services	2	4	2
Wholesale	16	5	11	Legal Services	3	5	5
Mechanical Eng	25	6	14	Mechanical Eng	16	6	16
Wood/Paper/Print	17	7	17	Agric.,Forestry	14	7	4
Energy/Water	11	8	25	Wholesale	7	8	21
Iron/Steel	31	9	21	Other Trans.	12	9	6
Postal System	23	10	24	Educ./Sport	8	10	14
Educ./Sport	21	11	22	Postal System	17	11	23
Agric.,Forestry	8	12	4	Energy/Water	23	12	25
Financial Inst	12	13	8	Health Service	5	13	17
Constr. Relate	6	14	5	Electrical Eng	22	14	20
Chemicals	22	15	30	Clothing/Text	20	15	29
Electrical Eng	26	16	28	Retail	30	16	7
Volunt./Church	24	17	20	Insurance	13	17	22
Other Trans.	15	18	16	Iron/Steel	11	18	19
Insurance	9	19	12	Food Industry	19	19	13
Train System	29	20	31	Synthetics	15	20	28
Restaurants	7	21	6	Construction	9	21	12
Retail	3	22	7	Chemicals	25	22	18
Earth/Clay/Stone	19	23	23	Service Indust	29	23	15
Synthetics	30	24	32	Earth/Clay/Stone	32	24	30
Social Security	27	25	15	Volunt./Church	26	25	31
Food Industry	13	26	13	Wood/Paper/Print	10	26	9
Service Indust	28	27	18	Train System	31	27	32
Other Services	2	28	3	Social Security	24	28	26
Health Service	10	29	19	Financial Inst	27	29	11
Legal Services	14	30	10	Restaurants	18	30	8
Mining	20	31	26	Mining	21	31	10
Fisheries	32	32	29	Fisheries	33	32	33
Priv. Househld	33	33	33	Priv. Househld	28	33	27

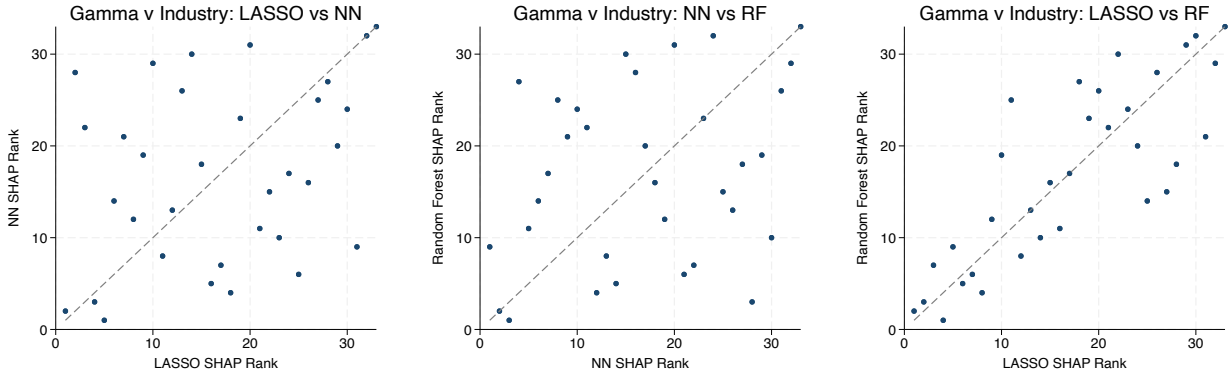


Figure 3: Gamma Industry SHAP Rank Comparisons (Hourly Earnings, flwage)

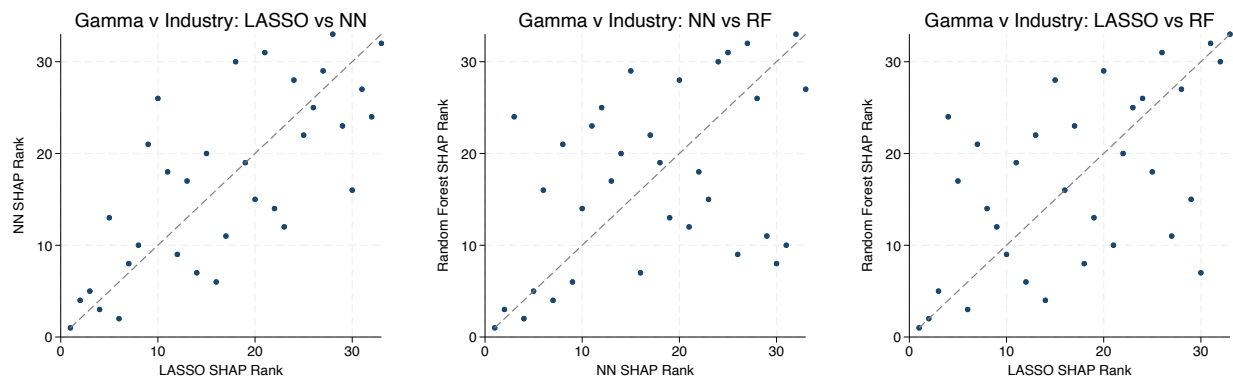


Figure 4: Gamma Industry SHAP Rank Comparisons (Annual Earnings, fearn)

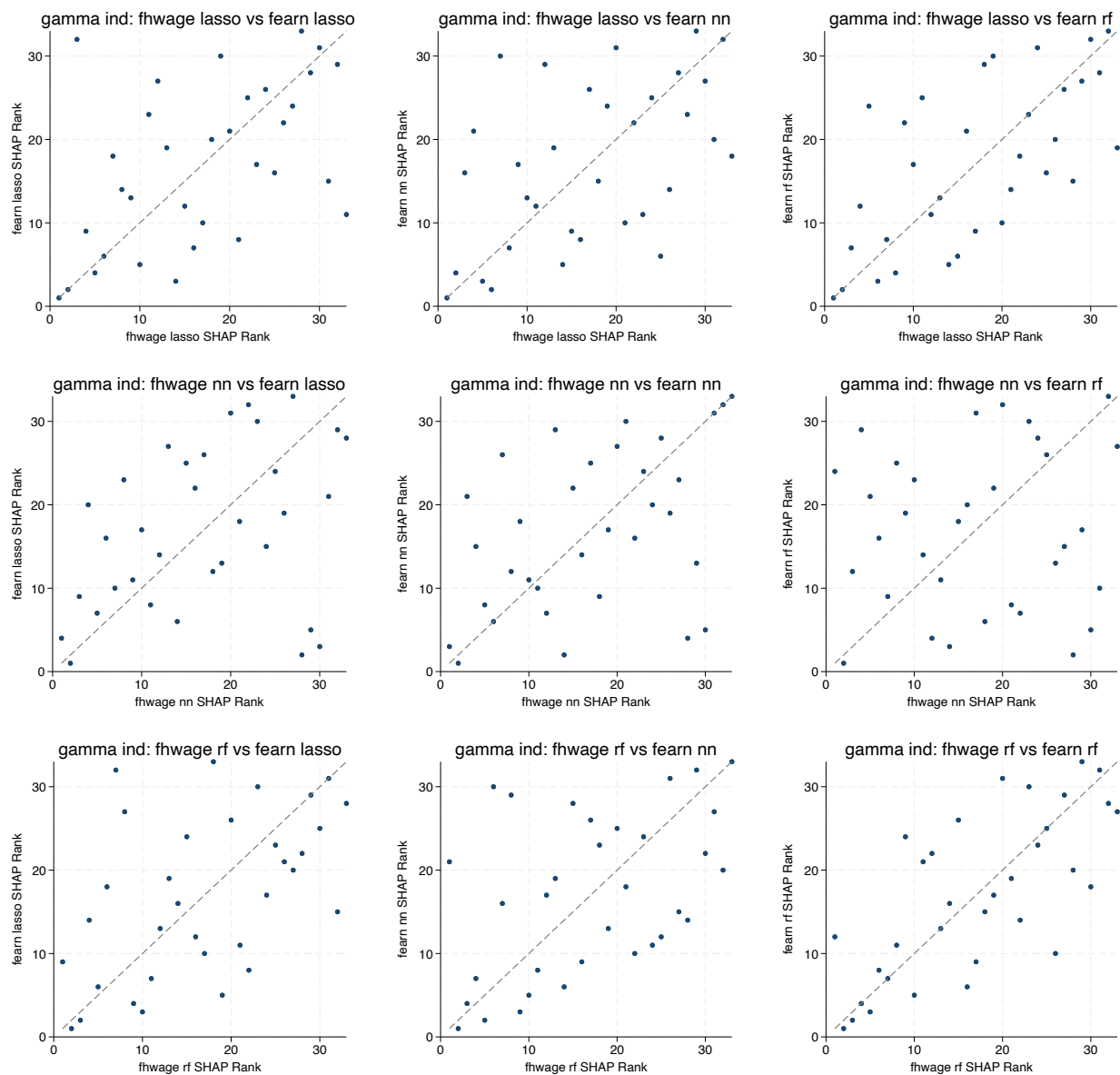


Figure 5: Gamma Industry SHAP Rank: fh wage (rows: LASSO, NN, RF) vs. fearn (columns: LASSO, NN, RF)

Table 4: Gamma Regressions: Occupation Selection Results

Hourly Earnings (flwage)				Annual Earnings (fearn)			
Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order	Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Labor/Craftsmn	10	1	19	Not Applicable	71	1	3
Transport.Oper	59	2	33	Transport.Oper	10	2	18
Convey. Oper.	4	3	6	Insurance Rep.	4	3	4
SecurityServic	56	4	52	Machine Fitter	78	4	13
Architect/Engineer	14	5	39	Priv.Bus.Leadr	1	5	1
Janitor	17	6	38	Farm Manager	2	6	2
Dr./Dentist/Vet	2	7	1	Educator	6	7	19
Tool/Die Maker	21	8	45	Electr. Enginr	5	8	23
Educator	9	9	16	Labor/Craftsmn	11	9	10
Inspector	16	10	31	Painter	14	10	8
Eng.Tech.Expert	25	11	54	Legislator	7	11	11
Legislator	15	12	11	Vendor	18	12	26
Vendor	6	13	4	Inspector	21	13	54
Electr. Enginr	13	14	41	Convey. Oper.	12	14	22
Mailman	41	15	58	Architect/Engineer	8	15	40
Ofc.Worker Etc	18	16	26	Mathematician	24	16	9
Scientist	7	17	37	Pipe Fitter	34	17	20
Tailor	57	18	48	Dr./Dentist/Vet	3	18	5
Not Applicable	8	19	8	Ofc.Worker Etc	67	19	24
Soldier	68	20	50	Accountant	17	20	36
Rest./StoreMgr	27	21	13	Janitor	20	21	14
Priv.Bus.Leadr	1	22	2	Scientist	32	22	50
Lawyer	23	23	17	Mailman	22	23	46
Pipe Fitter	55	24	25	Tool/Die Maker	9	24	30
Insurance Rep.	3	25	3	Bricklay/Carprt	69	25	12
Machine Fitter	69	26	14	RelatMedicalJob	13	26	45
RelatMedicalJob	51	27	60	Soldier	58	27	61
Mathematician	67	28	44	Eng.Tech.Expert	45	28	48
Hair Stylist	22	29	7	Rest./StoreMgr	25	29	7
Bricklay/Carprt	20	30	20	Lawyer	31	30	49
BusinessManagr	19	31	18	Tailor	39	31	57
Agriculturladm	37	32	22	BusinessManagr	16	32	29
Food Producer	49	33	51	Food Producer	26	33	56
Author	24	34	15	Economist	52	34	69
Sculptr/Paintr	39	35	53	Office Manager	62	35	43
Domestic Help	44	36	32	Bookkeepr/Cash	23	36	38
Cook/Waiter	12	37	24	SecurityServic	15	37	31
Life/Phys. Sci.	34	38	46	Cleric	38	38	53
Farm Manager	5	39	5	Printer Etc.	68	39	41
Service Worker	26	40	21	Buyer	28	40	17
Chemist	28	41	42	Administrator	49	41	63
Tel. Operator	31	42	9	Cook/Waiter	60	42	16
Stat.Mach.Oper	43	43	68	Forestry Work	72	43	72
Cabinet Maker	61	44	70	Cabinet Maker	36	44	65
Bookkeepr/Cash	63	45	49	ChemicalWorker	56	45	37
Prof. Athlete	33	46	28	Prof. Athlete	30	46	15
Painter	11	47	12	Lumbrman/Axman	35	47	6
Administrator	45	48	57	Life/Phys. Sci.	41	48	60
Lumbrman/Axman	29	49	10	Stat.Mach.Oper	54	49	71
Cleric	54	50	34	Foundry Worker	61	50	68
Office Manager	30	51	47	Chemist	33	51	58
ComputerOperat	38	52	66	Sculptr/Paintr	19	52	39
ChemicalWorker	71	53	61	Dry-Cleaner	47	53	35
Foundry Worker	47	54	67	ComputerOperat	55	54	59
Stenographer	48	55	73	Spinner/Weaver	48	55	74
Tech.Salespers	77	56	65	Transp. Attend	46	56	42
Aero/MarineEngr	32	57	23	Salesperson	65	57	52
Spinner/Weaver	46	58	75	Domestic Help	27	58	51
Accountant	50	59	36	Music/Perform	29	59	21
Dry-Cleaner	60	60	71	Author	53	60	32
Buyer	36	61	29	Aero/MarineEngr	44	61	55
Farm Hand	76	62	27	Tel. Operator	37	62	33
Economist	42	63	30	Service Worker	42	63	44
Miner	73	64	43	Broadcaster	77	64	76
Salesperson	72	65	63	Miner	76	65	47
Broadcaster	78	66	69	Tech.Salespers	43	66	28
Transp. Attend	40	67	40	Fisher/Hunter	50	67	27
Forestry Work	74	68	64	Hair Stylist	40	68	34
Shoemaker	65	69	77	Agriculturladm	73	69	64
Jewelry Maker	52	70	62	Manufacturer	63	70	77
Glazier	62	71	78	Shoemaker	59	71	67
Music/Perform	35	72	35	Stenographer	70	72	66
Fisher/Hunter	75	73	59	Glazier	64	73	78
Printer Etc.	53	74	56	Farm Hand	51	74	70
HH Supervisor	70	75	74	Jewelry Maker	75	75	25
Manufacturer	64	76	76	Conductor	74	76	75
Conductor	66	77	55	Stone Cutter	57	77	62
Stone Cutter	58	78	72	HH Supervisor	66	78	73

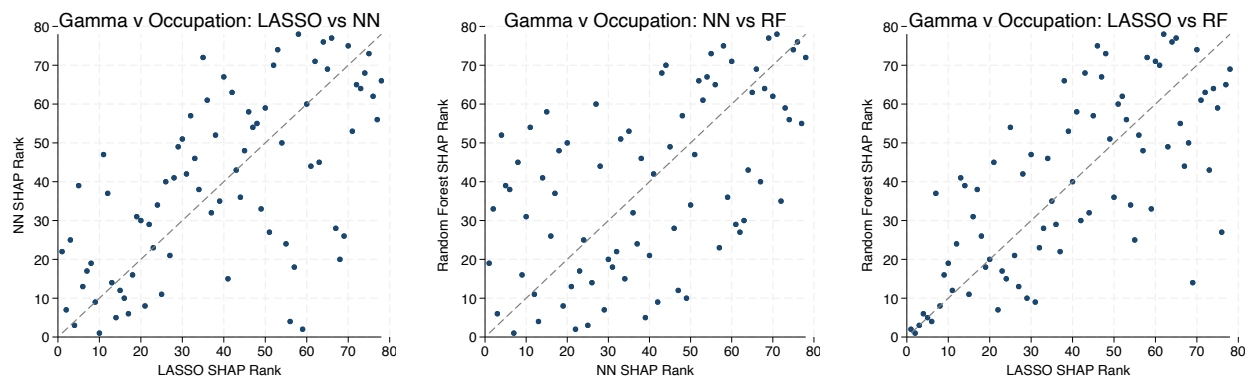


Figure 6: Gamma Occupation SHAP Rank Comparisons (Hourly Earnings, fhwage)

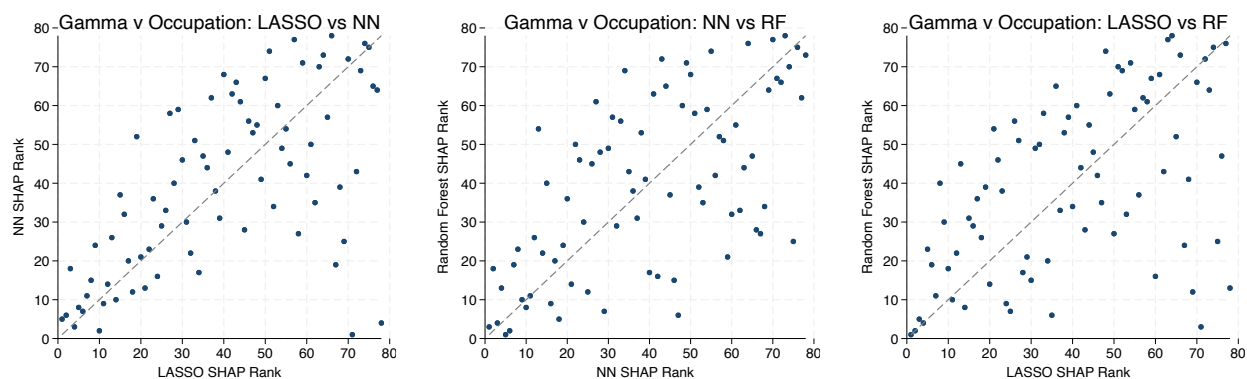


Figure 7: Gamma Occupation SHAP Rank Comparisons (Annual Earnings, fearn)

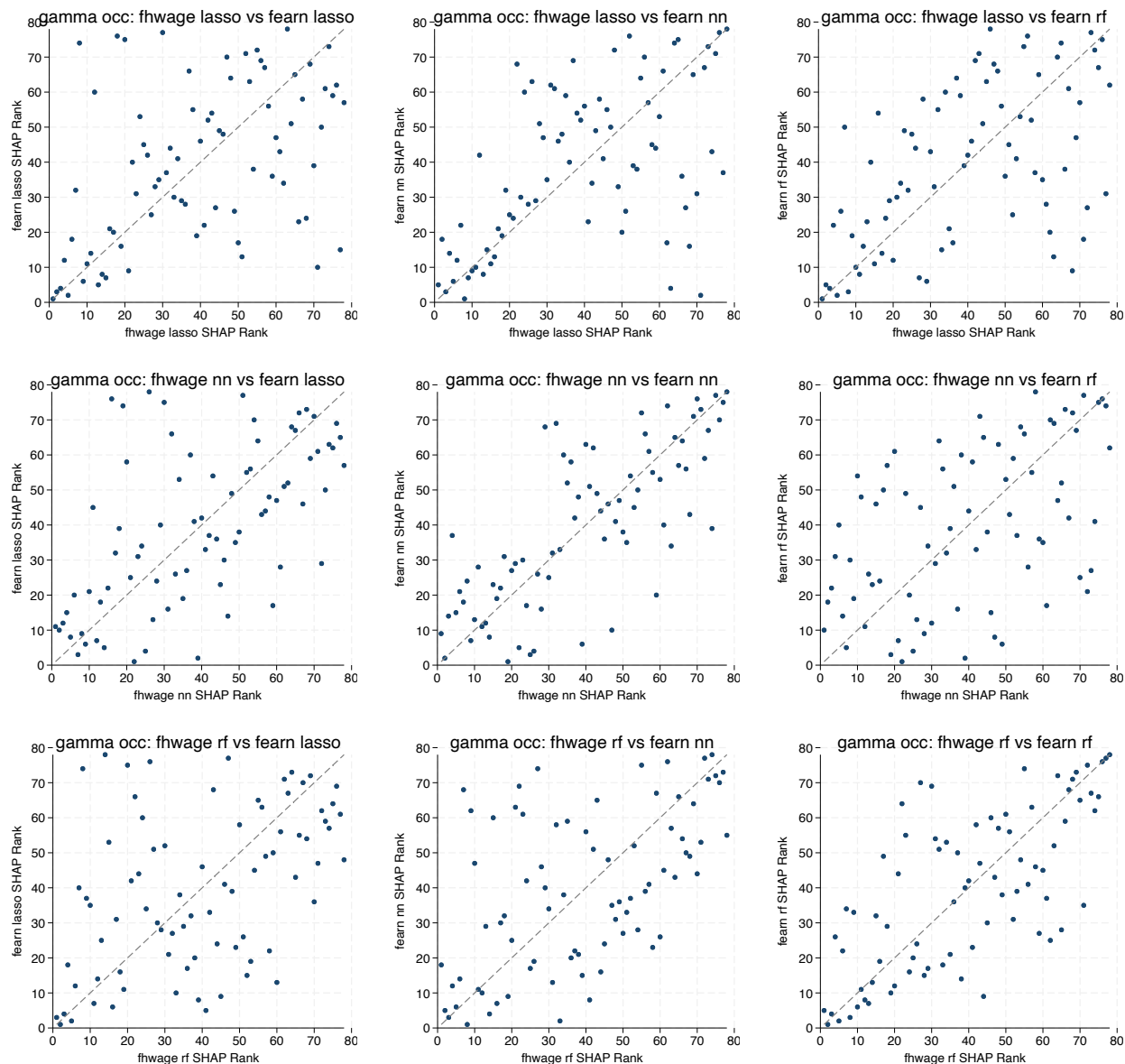


Figure 8: Gamma Occupation SHAP Rank: fh wage (rows: LASSO, NN, RF) vs. fearn (columns: LASSO, NN, RF)

Table 5: Gamma Regressions: OLS Results (Coefficients x 100), Hourly Earnings (flwage)

	(1)	(2)	(3)	(4)	(5)
	Gamma (flwage)	Gamma (flwage)	Gamma (flwage)	Gamma (flwage)	Gamma (flwage)
High School Graduate	-0.0607 (0.114)	-0.00350 (0.120)	0.0445 (0.121)	0.0233 (0.121)	0.0427 (0.122)
Some College	0.279* (0.144)	0.342** (0.151)	0.382** (0.153)	0.318** (0.156)	0.347** (0.157)
Bachelor's Degree	0.348** (0.142)	0.405*** (0.150)	0.453*** (0.156)	0.310* (0.163)	0.355** (0.164)
Bachelors+	0.0674 (0.176)	0.139 (0.188)	0.193 (0.197)	0.0192 (0.209)	0.0476 (0.211)
22-29	0.000486 (0.00136)	0.00511** (0.00203)	0.00470** (0.00203)	0.00512** (0.00203)	0.00496** (0.00204)
30-37	0.000842 (0.00122)	0.00424*** (0.00160)	0.00404** (0.00160)	0.00441*** (0.00160)	0.00427*** (0.00160)
38-45	-0.00113 (0.00125)	0.000550 (0.00135)	0.000448 (0.00135)	0.000742 (0.00135)	0.000642 (0.00135)
54-61	-0.000658 (0.00157)	-0.00148 (0.00161)	-0.00173 (0.00161)	-0.00184 (0.00161)	-0.00194 (0.00161)
62-69	0.0110*** (0.00334)	0.00935*** (0.00340)	0.00842** (0.00340)	0.00819** (0.00341)	0.00767** (0.00341)
Probability of Recession	-0.000325 (0.00223)	-0.305 (5.103)	1.186 (5.148)	1.417 (5.135)	1.707 (5.158)
5-year moving average of AEP	0.00258 (0.00203)	0.00302 (0.00218)	0.000203 (0.00229)	0.00381 (0.00234)	0.00251 (0.00240)
Out of Labor Force	-0.555 (0.373)	-0.560 (0.374)	-0.716* (0.378)	-0.708* (0.378)	-0.745** (0.378)
Tenure	0.00405 (0.00602)	0.00687 (0.00658)	0.00897 (0.00666)	0.00635 (0.00669)	0.00738 (0.00671)
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes
R-squared	0.000	0.001	0.002	0.004	0.005
N	79633	79633	79633	79633	79633

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Gamma Regressions: OLS Results (Coefficients x 100), Annual Earnings (fearn)

	(1)	(2)	(3)	(4)	(5)
	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)
High School Graduate	0.0995 (0.167)	0.159 (0.174)	0.227 (0.176)	0.210 (0.177)	0.244 (0.177)
Some College	0.488** (0.209)	0.569*** (0.219)	0.607*** (0.222)	0.508** (0.227)	0.547** (0.228)
Bachelor's Degree	0.410** (0.207)	0.496** (0.219)	0.483** (0.226)	0.311 (0.236)	0.368 (0.238)
Bachelors+	0.251 (0.257)	0.409 (0.273)	0.335 (0.288)	0.228 (0.305)	0.247 (0.308)
22-29	-0.000430 (0.00197)	0.00237 (0.00295)	0.00141 (0.00296)	0.00218 (0.00296)	0.00195 (0.00296)
30-37	0.00132 (0.00176)	0.00360 (0.00232)	0.00326 (0.00232)	0.00385* (0.00232)	0.00365 (0.00233)
38-45	-0.000776 (0.00182)	0.000591 (0.00196)	0.000454 (0.00196)	0.000847 (0.00196)	0.000752 (0.00196)
54-61	0.00418* (0.00240)	0.00364 (0.00246)	0.00303 (0.00246)	0.00299 (0.00246)	0.00267 (0.00246)
62-69	0.0303*** (0.00572)	0.0293*** (0.00583)	0.0281*** (0.00583)	0.0281*** (0.00584)	0.0273*** (0.00584)
Probability of Recession	0.00198 (0.00325)	-7.459 (7.519)	-2.728 (7.572)	-2.879 (7.557)	-1.827 (7.589)
5-year moving average of AEP	0.0169*** (0.00293)	0.0195*** (0.00314)	0.0141*** (0.00331)	0.0182*** (0.00337)	0.0168*** (0.00346)
Out of Labor Force	-0.356 (0.595)	-0.435 (0.595)	-0.931 (0.601)	-0.857 (0.601)	-0.956 (0.601)
Tenure	0.0000583 (0.00895)	0.00544 (0.00977)	0.00881 (0.00992)	0.00706 (0.00996)	0.00767 (0.00999)
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes
R-squared	0.001	0.002	0.004	0.005	0.005
N	78216	78216	78216	78216	78216

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

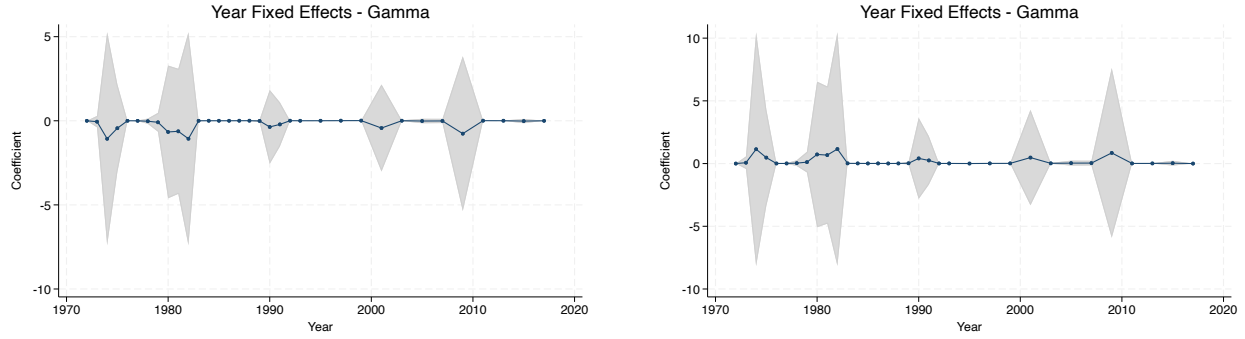


Figure 9: Year Fixed Effects on Gamma (All-Controls Model, 95% CI): Hourly earnings (fhwage, left) and Annual earnings (fearn, right)

Table 7: Gamma Regressions: Stepwise Selection Results (Coefficients x 100), Hourly Earnings (fhwage)

	(1)	(2)	(3)	(4)	(5)
	Gamma (fhwage)	Gamma (fhwage)	Gamma (fhwage)	Gamma (fhwage)	Gamma (fhwage)
Age 62-69	0.0110*** (0.00319)	0.0109*** (0.00320)	0.00979** (0.00320)	0.00943** (0.00320)	0.00897** (0.00320)
Some College	0.295** (0.112)	0.300** (0.113)	0.307** (0.113)	0.282* (0.115)	0.281* (0.116)
Bachelor's Degree	0.338** (0.103)	0.345*** (0.104)	0.368*** (0.106)	0.277* (0.114)	0.276* (0.114)
5-year moving average of AEP				0.00407* (0.00203)	
Constant	0.0139*** (0.000473)	0.0134*** (0.00113)	0.00898*** (0.00153)	0.00968** (0.00350)	0.0112** (0.00340)
Census Division FE		✓			
Year FE					
Race FE					
Cohort FE					
Occupation FE				✓	✓
Industry FE			✓		✓
Census Division FE Available	No	Yes	Yes	Yes	Yes
Year FE Available	No	Yes	Yes	Yes	Yes
Race FE Available	No	Yes	Yes	Yes	Yes
Cohort FE Available	No	Yes	Yes	Yes	Yes
Occupation FE Available	No	No	No	Yes	Yes
Industry FE Available	No	No	Yes	No	Yes
R-squared	0.000	0.001	0.002	0.003	0.004
N	79633	79633	79633	79633	79633

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 8: Gamma Regressions: Stepwise Selection Results (Coefficients x 100), Annual Earnings (fearn)

	(1)	(2)	(3)	(4)	(5)
	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)
5-year moving average of AEP	0.0142*** (0.00250)	0.0145*** (0.00256)	0.00941*** (0.00274)	0.0156*** (0.00299)	0.0139*** (0.00308)
Some College	0.318* (0.160)				
Age 62-69	0.0299*** (0.00555)	0.0298*** (0.00555)	0.0285*** (0.00556)	0.0282*** (0.00557)	0.0275*** (0.00557)
Probability of Recession			0.0149* (0.00749)	0.0158* (0.00749)	0.0154* (0.00749)
Constant	0.0156*** (0.00116)	0.0168*** (0.00182)	0.0125*** (0.00296)	0.00941 (0.00539)	0.01000 (0.00540)
Census Division FE		✓	✓	✓	✓
Year FE			✓	✓	✓
Race FE					
Cohort FE					
Occupation FE				✓	✓
Industry FE			✓		✓
Census Division FE Available	No	Yes	Yes	Yes	Yes
Year FE Available	No	Yes	Yes	Yes	Yes
Race FE Available	No	Yes	Yes	Yes	Yes
Cohort FE Available	No	Yes	Yes	Yes	Yes
Occupation FE Available	No	No	No	Yes	Yes
Industry FE Available	No	No	Yes	No	Yes
R-squared	0.001	0.001	0.003	0.004	0.005
N	78216	78216	78216	78216	78216

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 9: Gamma Regressions: Lasso Selection Results, Hourly Earnings (flwage)

	No Controls	No Occ/Ind	No Occ	No Ind	All Controls
EDU1	2	4	10	11	11
EDU2	4	5	3	3	3
EDU3	3	3	2	5	4
EDU4	11	13	11	12	12
OLF	5	8	4	5	4
PrRecess	12				
agecat1	10	9	8	8	7
agecat2	6	7	5	4	4
agecat3	4	6	6	5	5
agecat5	8	10	7	6	6
agecat6	1	2	2	2	2
ma5aep	7	11	14	7	8
tenure	9	12	9	9	9
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes

Table 10: Gamma Regressions: Lasso Selection Results, Annual Earnings (fearn)

	No Controls	No Occ/Ind	No Occ	No Ind	All Controls
EDU1	10	12	11	10	9
EDU2	2	3	4	4	4
EDU3	5	6	9	9	8
EDU4	8	8	12	12	11
OLF	7	9	6	6	6
PrRecess	7				
agecat1	9	10	13	11	10
agecat2	6	5	5	5	5
agecat3	4	4	7	8	7
agecat5	3	7	8	7	7
agecat6	1	2	2	2	2
ma5aep	1	2	3	3	3
tenure		11	10	10	9
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes

5.2 Transitory Shocks

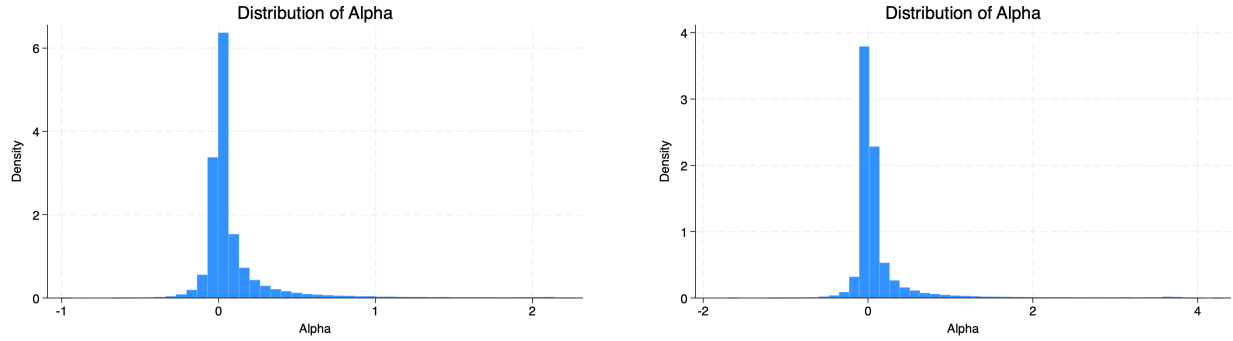


Figure 10: Distribution of Alpha: Hourly earnings (flwage, left) and Annual earnings (fearn, right)

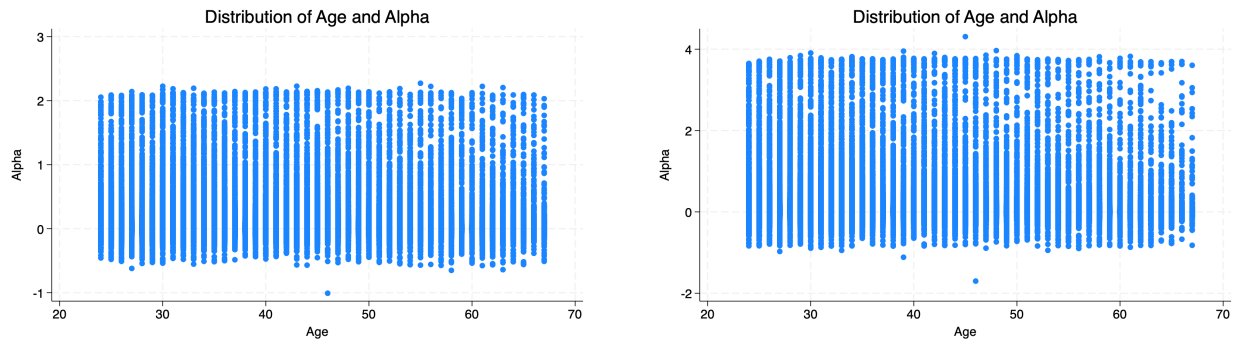


Figure 11: Scatterplot of Age vs. Alpha: Hourly earnings (flwage, left) and Annual earnings (fearn, right)

Table 11: Alpha Regressions: F-Test Results (Hourly Earnings, flwage)

	No Occ/Ind (1)	No Occ (2)	No Ind (3)	All Controls (4)
Census Division FE	14.41*** (0.0000)	7.94*** (0.0000)	9.02*** (0.0000)	7.28*** (0.0000)
Year FE	12.77*** (0.0000)	14.61*** (0.0000)	14.10*** (0.0000)	14.31*** (0.0000)
Race FE	4.15*** (0.0009)	4.36*** (0.0006)	4.95*** (0.0002)	5.23*** (0.0001)
Cohort FE	1.66 (0.1724)	2.98** (0.0300)	2.32* (0.0734)	2.95** (0.0314)
Occupation FE	—	—	17.49*** (0.0000)	7.04*** (0.0000)
Industry FE	—	35.65*** (0.0000)	—	10.55*** (0.0000)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

Table 12: Alpha Regressions: F-Test Results (Annual Earnings, fearn)

	No Occ/Ind (1)	No Occ (2)	No Ind (3)	All Controls (4)
Census Division FE	11.18*** (0.0000)	7.43*** (0.0000)	8.76*** (0.0000)	7.67*** (0.0000)
Year FE	8.19*** (0.0000)	15.35*** (0.0000)	14.37*** (0.0000)	15.11*** (0.0000)
Race FE	6.35*** (0.0000)	7.54*** (0.0000)	9.06*** (0.0000)	9.25*** (0.0000)
Cohort FE	1.05 (0.3710)	2.26* (0.0791)	1.88 (0.1306)	2.19* (0.0871)
Occupation FE	—	—	17.65*** (0.0000)	4.76*** (0.0000)
Industry FE	—	39.58*** (0.0000)	—	8.58*** (0.0000)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

Table 13: Alpha Regressions: Lasso Industry Selection Results

Hourly Earnings (flwage)

Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Not Applicable	1	1	1
Other Services	2	2	2
Public Administration	6	3	16
Agric./Forestry	4	4	3
Mechanical Eng	12	5	19
Retail	7	6	8
Legal Services	9	7	4
Educ./Sport	11	8	21
Chemicals	22	9	29
Iron/Steel	14	10	27
Electrical Eng	25	11	22
Food Industry	33	12	14
Wood/Paper/Print	13	13	9
Constr. Relate	3	14	5
Health Service	17	15	18
Postal System	28	16	24
Energy/Water	23	17	28
Clothing/Text	15	18	31
Wholesale	24	19	12
Synthetics	21	20	32
Construction	5	21	6
Restaurants	31	22	20
Insurance	18	23	26
Volunt./Church	27	24	23
Train System	20	25	30
Service Indust	8	26	10
Fisheries	26	27	15
Earth/Clay/Stone	32	28	25
Other Trans.	10	29	7
Mining	16	30	13
Financial Inst	19	31	11
Priv. Household	30	32	33
Social Security	29	33	17

Annual Earnings (fearn)

Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Not Applicable	1	1	1
Other Services	2	2	2
Public Administration	6	3	7
Mechanical Eng	26	4	16
Retail	7	5	8
Educ./Sport	9	6	24
Agric./Forestry	4	7	5
Electrical Eng	14	8	27
Health Service	17	9	22
Chemicals	11	10	31
Postal System	15	11	25
Energy/Water	16	12	23
Insurance	12	13	29
Other Trans.	10	14	11
Iron/Steel	33	15	17
Wood/Paper/Print	24	16	12
Financial Inst	32	17	10
Wholesale	19	18	14
Clothing/Text	20	19	20
Construction	13	20	6
Legal Services	5	21	4
Synthetics	18	22	30
Constr. Relate	3	23	3
Food Industry	31	24	15
Service Indust	23	25	9
Restaurants	8	26	13
Social Security	27	27	32
Mining	22	28	26
Volunt./Church	28	29	18
Train System	29	30	19
Earth/Clay/Stone	21	31	21
Fisheries	25	32	28
Priv. Household	30	33	33

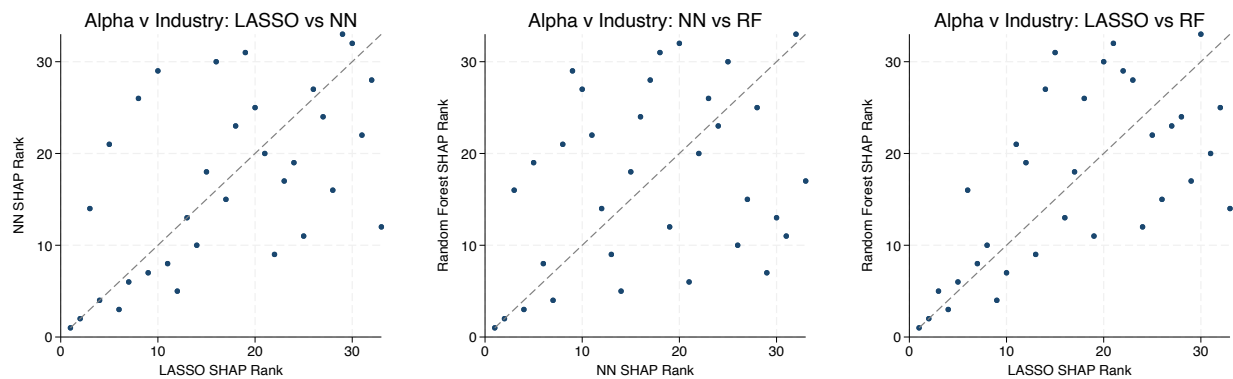


Figure 12: Alpha Industry SHAP Rank Comparisons (Hourly Earnings, fhwage)

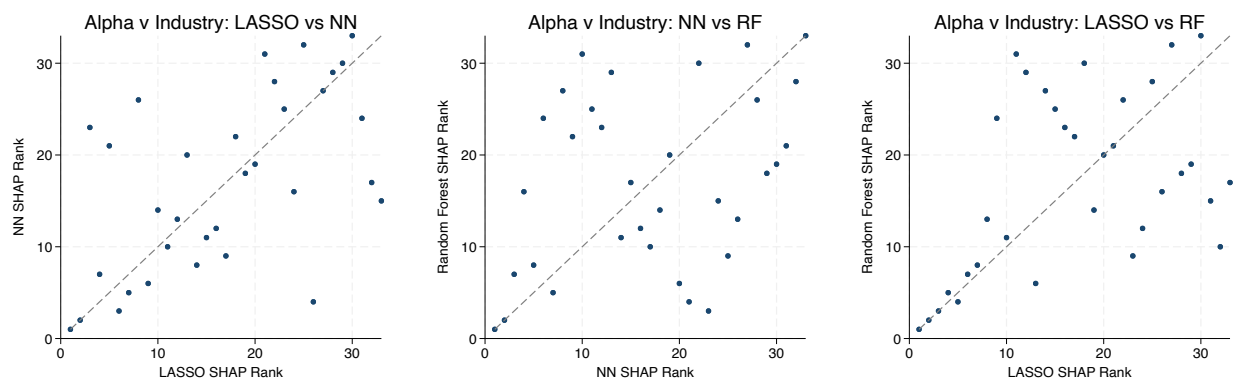


Figure 13: Alpha Industry SHAP Rank Comparisons (Annual Earnings, fearn)

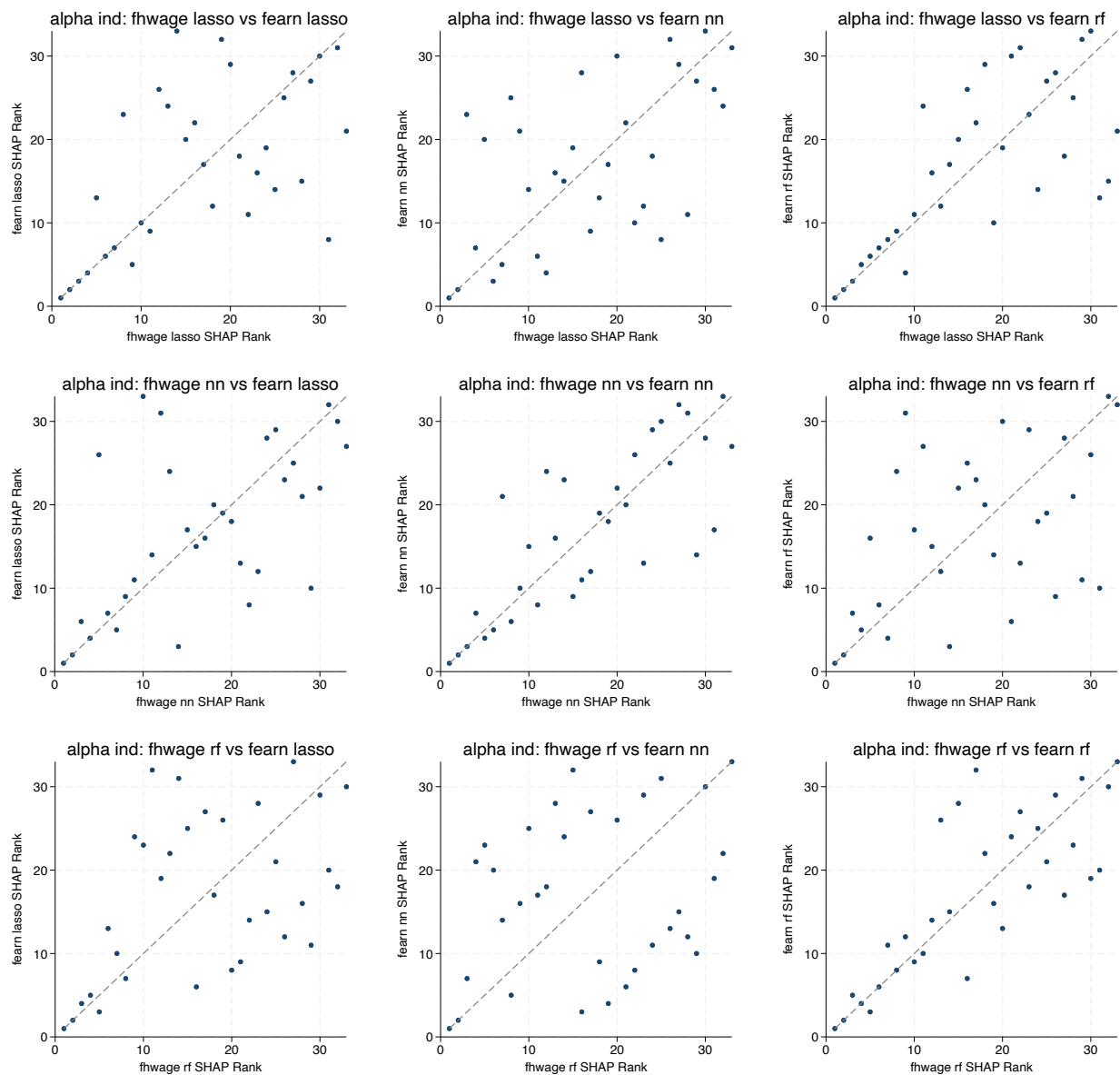


Figure 14: Alpha Industry SHAP Rank: fh wage (rows: LASSO, NN, RF) vs. fearn (columns: LASSO, NN, RF)

Table 14: Alpha Regressions: Lasso Occupation Selection Results

Hourly Earnings (flhwage)				Annual Earnings (fearn)			
Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order	Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Not Applicable	1	1	1	Not Applicable	1	1	1
Farm Manager	2	2	2	Farm Manager	2	2	2
Architect/Engineer	6	3	18	Machine Fitter	13	3	8
Machine Fitter	7	4	11	Insurance Rep.	5	4	3
Priv.Bus.Leadr	3	5	3	Inspector	7	5	45
Mathematician	5	6	30	Architect/Engineer	8	6	32
Inspector	4	7	41	SecurityServic	25	7	18
Electr. Enginr	24	8	26	Mathematician	4	8	33
Transport. Oper	29	9	7	Transport. Oper	11	9	4
Convey. Oper.	21	10	32	BusinessManagr	21	10	37
Mailman	16	11	35	Mailman	14	11	24
Janitor	17	12	24	Janitor	20	12	20
SecurityServic	71	13	28	Educator	18	13	26
Insurance Rep.	10	14	4	Electr. Enginr	47	14	12
Ofc.Worker Etc	9	15	36	Eng.Tech.Expert	12	15	40
Bricklay/Carpt	30	16	5	Ofc.Worker Etc	34	16	21
BusinessManagr	15	17	46	Labor/Craftsmn	6	17	5
Labor/Craftsmn	20	18	8	Vendor	22	18	17
Eng.Tech.Expert	12	19	50	Convey. Oper.	32	19	13
Rest./StoreMgr	13	20	9	Scientist	15	20	28
RelatMedicalJob	59	21	33	Soldier	73	21	48
Stat.Mach.Oper	43	22	66	Food Producer	19	22	57
Office Manager	11	23	59	Priv.Bus.Leadr	3	23	6
Scientist	38	24	27	Accountant	27	24	36
Educator	18	25	17	Tailor	66	25	41
Lumbrman/Axman	34	26	15	Hair Stylist	26	26	15
Accountant	42	27	29	Lumbrman/Axman	23	27	9
Tool/Die Maker	35	28	55	Aero/MarineEngr	30	28	66
Printer Etc.	31	29	54	Office Manager	35	29	54
Food Producer	23	30	49	Printer Etc.	68	30	38
Bookkeepr/Cash	46	31	40	ComputerOperat	36	31	71
Hair Stylist	40	32	22	Lawyer	40	32	44
Tel. Operator	52	33	63	Foundry Worker	60	33	52
Service Worker	54	34	56	Bookkeepr/Cash	59	34	27
Vendor	28	35	13	Tool/Die Maker	43	35	50
Lawyer	45	36	21	Painter	28	36	14
Cook/Waiter	25	37	14	Bricklay/Carpt	10	37	7
Fisher/Hunter	37	38	23	Cleric	38	38	30
Aero/MarineEngr	32	39	10	Chemist	70	39	55
Soldier	14	40	25	Rest./StoreMgr	39	40	22
Life/Phys. Sci.	26	41	34	RelatMedicalJob	33	41	16
Administrator	67	42	58	Cook/Waiter	16	42	11
ComputerOperat	41	43	78	Tel. Operator	51	43	58
Sculptr/Paintr	53	44	37	Music/Perform	31	44	25
Tailor	44	45	51	Legislator	17	45	19
ChemicalWorker	65	46	62	Stat.Mach.Oper	45	46	60
Cleric	39	47	53	Author	62	47	42
Stenographer	50	48	74	Administrator	57	48	63
Economist	63	49	38	Pipe Fitter	9	49	10
Pipe Fitter	27	50	16	Broadcaster	74	50	73
Music/Perform	33	51	6	Salesperson	61	51	62
Broadcaster	69	52	52	Domestic Help	63	52	35
Prof. Athlete	47	53	42	Fisher/Hunter	48	53	43
Salesperson	58	54	67	Spinner/Weaver	52	54	65
Painter	22	55	39	Prof. Athlete	42	55	46
Cabinet Maker	64	56	61	Forestry Work	54	56	68
Miner	55	57	43	Sculptr/Paintr	49	57	31
Buyer	60	58	68	ChemicalWorker	64	58	59
Spinner/Weaver	49	59	60	Miner	78	59	61
Foundry Worker	68	60	57	Service Worker	50	60	23
Dr./Dentist/Vet	8	61	20	Cabinet Maker	24	61	49
Legislator	19	62	12	Stenographer	53	62	69
Glazier	57	63	71	Life/Phys. Sci.	46	63	39
Domestic Help	72	64	47	Dry-Cleaner	37	64	51
Author	62	65	31	Glazier	55	65	76
Tech.Salespers	51	66	48	Economist	72	66	53
Transp. Attend	76	67	73	Agriculturladm	44	67	29
Chemist	70	68	19	Buyer	67	68	56
Forestry Work	48	69	65	Jewelry Maker	65	69	72
Conductor	77	70	77	Shoemaker	56	70	70
Dry-Cleaner	56	71	44	Dr./Dentist/Vet	29	71	47
Manufacturer	75	72	70	Farm Hand	58	72	64
Shoemaker	73	73	75	Transp. Attend	77	73	67
Farm Hand	78	74	64	Manufacturer	71	74	78
Agriculturladm	36	75	45	Stone Cutter	69	75	77
HH Supervisor	66	76	72	HH Supervisor	75	76	74
Stone Cutter	74	77	76	Tech.Salespers	41	77	34
Jewelry Maker	61	78	69	Conductor	76	78	75

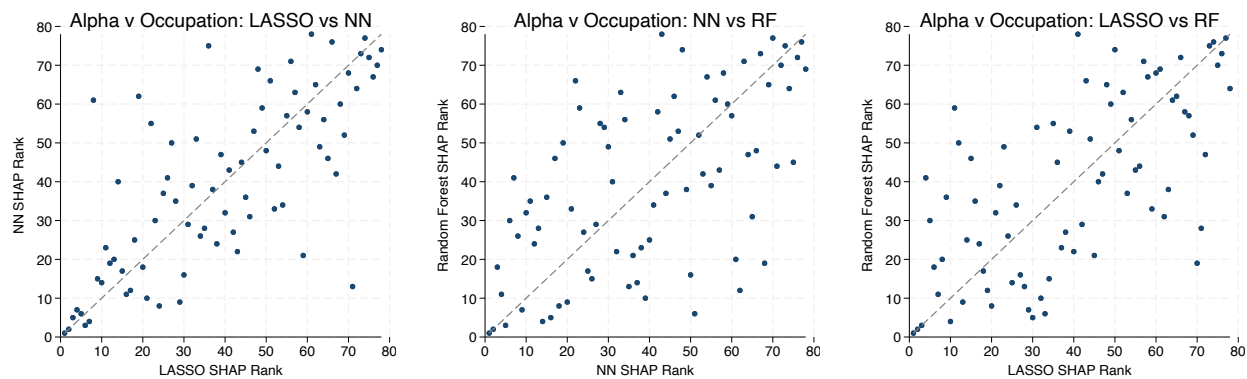


Figure 15: Alpha Occupation SHAP Rank Comparisons (Hourly Earnings, fh wage)

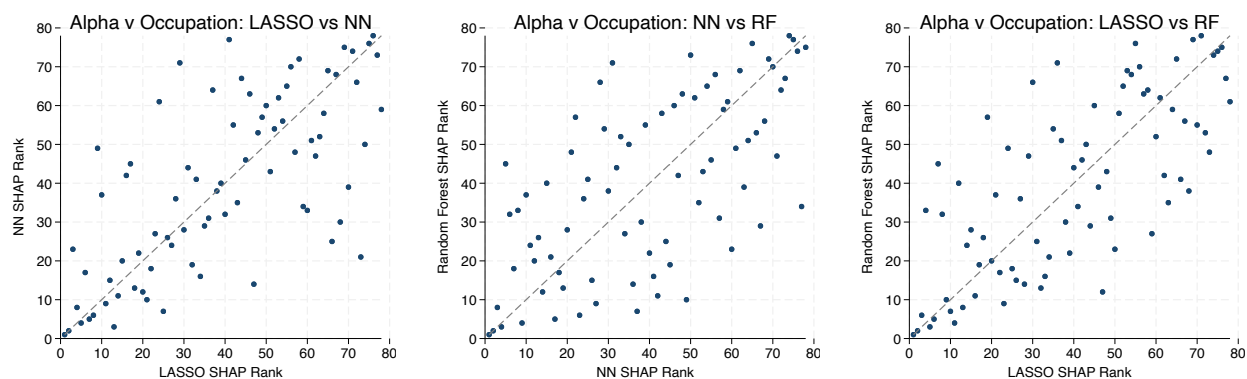


Figure 16: Alpha Occupation SHAP Rank Comparisons (Annual Earnings, fearn)

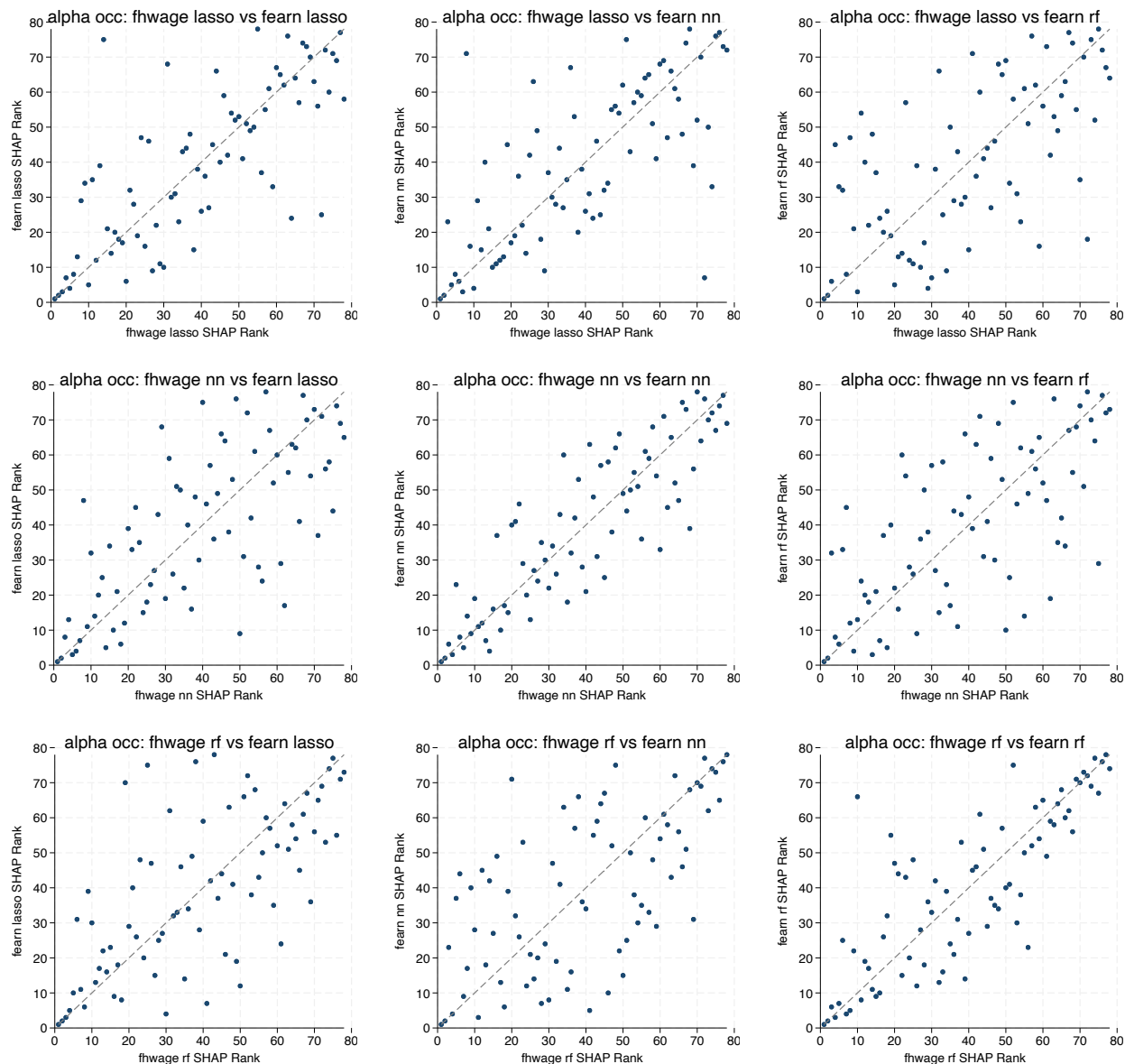


Figure 17: Alpha Occupation SHAP Rank: fh wage (rows: LASSO, NN, RF) vs. fearn (columns: LASSO, NN, RF)

Table 15: Alpha Regressions: OLS Results (Coefficients x 100), Hourly Earnings (flwage)

	(1)	(2)	(3)	(4)	(5)
	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)
High School Graduate	0.764*** (0.254)	0.409 (0.265)	0.920*** (0.266)	0.858*** (0.267)	0.994*** (0.267)
Some College	1.374*** (0.322)	1.021*** (0.337)	1.527*** (0.340)	1.233*** (0.346)	1.390*** (0.347)
Bachelor's Degree	1.648*** (0.319)	1.416*** (0.337)	1.898*** (0.347)	1.069*** (0.362)	1.340*** (0.364)
Bachelors+	2.387*** (0.395)	2.231*** (0.419)	2.842*** (0.439)	1.752*** (0.465)	1.993*** (0.468)
22-29	-0.0211*** (0.00310)	-0.0126*** (0.00452)	-0.0143*** (0.00450)	-0.0131*** (0.00450)	-0.0134*** (0.00450)
30-37	-0.0125*** (0.00277)	-0.00635* (0.00357)	-0.00647* (0.00356)	-0.00574 (0.00356)	-0.00596* (0.00356)
38-45	-0.00285 (0.00285)	-0.000775 (0.00306)	-0.000951 (0.00304)	-0.000553 (0.00304)	-0.000733 (0.00304)
54-61	0.00194 (0.00345)	0.00366 (0.00353)	0.00174 (0.00352)	0.00206 (0.00352)	0.00154 (0.00352)
62-69	0.0481*** (0.00582)	0.0503*** (0.00599)	0.0447*** (0.00597)	0.0460*** (0.00597)	0.0436*** (0.00597)
Probability of Recession	0.00973* (0.00514)	-0.996 (0.957)	1.060 (0.961)	1.331 (0.958)	1.524 (0.961)
5-year moving average of AEP	0.137*** (0.00447)	0.142*** (0.00479)	0.116*** (0.00502)	0.129*** (0.00511)	0.125*** (0.00524)
Out of Labor Force	9.787*** (0.744)	9.666*** (0.744)	7.546*** (0.752)	7.549*** (0.750)	7.336*** (0.751)
Tenure	-0.0884*** (0.0132)	-0.0124 (0.0146)	-0.0121 (0.0147)	-0.0196 (0.0147)	-0.0142 (0.0147)
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes
R-squared	.0162825	.0220699	.0331621	.0351438	.0384102
N	99542	99542	99542	99542	99542

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 16: Alpha Regressions: OLS Results (Coefficients x 100), Annual Earnings (fearn)

	(1)	(2)	(3)	(4)	(5)
	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)
High School Graduate	0.930** (0.437)	0.299 (0.457)	0.981** (0.458)	0.767* (0.460)	1.004** (0.461)
Some College	1.061* (0.554)	0.564 (0.580)	1.327** (0.584)	0.621 (0.596)	0.944 (0.597)
Bachelor's Degree	3.171*** (0.550)	2.855*** (0.581)	3.933*** (0.596)	2.219*** (0.623)	2.751*** (0.627)
Bachelors+	4.649*** (0.682)	4.375*** (0.723)	5.808*** (0.757)	3.582*** (0.801)	4.085*** (0.807)
22-29	-0.0311*** (0.00530)	-0.0184** (0.00780)	-0.0255*** (0.00776)	-0.0229*** (0.00777)	-0.0243*** (0.00777)
30-37	-0.0187*** (0.00474)	-0.0112* (0.00615)	-0.0116* (0.00612)	-0.0100 (0.00612)	-0.0105* (0.00612)
38-45	-0.00216 (0.00489)	0.000715 (0.00524)	0.000301 (0.00521)	0.00158 (0.00521)	0.00107 (0.00521)
54-61	-0.0000151 (0.00610)	-0.000685 (0.00625)	-0.00527 (0.00621)	-0.00488 (0.00622)	-0.00559 (0.00621)
62-69	0.0481*** (0.0115)	0.0467*** (0.0119)	0.0395*** (0.0118)	0.0402*** (0.0118)	0.0369*** (0.0118)
Probability of Recession	0.0693*** (0.00885)	0.195 (1.671)	5.649*** (1.674)	5.343*** (1.670)	6.085*** (1.675)
5-year moving average of AEP	0.381*** (0.00763)	0.397*** (0.00819)	0.351*** (0.00859)	0.373*** (0.00874)	0.371*** (0.00897)
Out of Labor Force	27.85*** (1.437)	27.21*** (1.436)	20.56*** (1.445)	20.97*** (1.443)	20.17*** (1.444)
Tenure	-0.194*** (0.0232)	-0.0949*** (0.0256)	-0.0432* (0.0258)	-0.0465* (0.0259)	-0.0410 (0.0259)
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes
R-squared	.0383024	.0424165	.0546593	.0555492	.0581946
N	97886	97886	97886	97886	97886

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

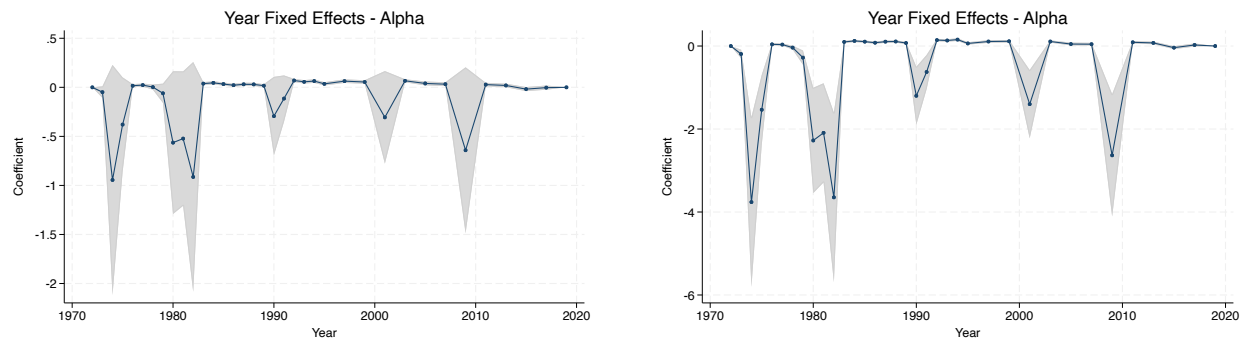


Figure 18: Year Fixed Effects on Alpha (All-Controls Model, 95% CI): Hourly earnings (fh wage, left) and Annual earnings (fearn, right)

Table 17: Alpha Regressions: Stepwise Selection Results (Coefficients x 100), Hourly Earnings (flwage)

	(1)	(2)	(3)	(4)	(5)
	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)
High School Graduate	0.708** (0.253)		0.727** (0.265)		0.809** (0.266)
Some College	1.305*** (0.320)	0.525* (0.264)	1.204*** (0.338)		1.076** (0.346)
Bachelor's Degree	1.593*** (0.319)	0.849** (0.258)	1.496*** (0.345)		0.924* (0.363)
Bachelors+	2.298*** (0.392)	1.465*** (0.353)	2.240*** (0.436)		1.345** (0.465)
Age 62-69	0.0488*** (0.00555)	0.0424*** (0.00569)	0.0373*** (0.00566)	0.0392*** (0.00566)	0.0365*** (0.00566)
5-year moving average of AEP	0.137*** (0.00446)	0.138*** (0.00462)	0.111*** (0.00495)	0.122*** (0.00483)	0.120*** (0.00517)
Out of Labor Force	9.840*** (0.743)	9.840*** (0.743)	7.741*** (0.752)	7.778*** (0.750)	7.517*** (0.750)
Tenure	-0.0842*** (0.0130)	-0.0340* (0.0137)	-0.0322* (0.0139)	-0.0431** (0.0139)	-0.0351* (0.0139)
Age 22-29	-0.0197*** (0.00252)				
Age 30-37	-0.0115*** (0.00212)				
Probability of Recession		0.0337** (0.0116)	0.0325** (0.0115)	0.0324** (0.0115)	0.0322** (0.0115)
Constant	0.0371*** (0.00332)	0.0585*** (0.0121)	0.0314* (0.0129)	0.0686*** (0.0142)	0.0556*** (0.0146)
Census Division FE		✓	✓	✓	✓
Year FE		✓	✓	✓	✓
Race FE		✓	✓	✓	✓
Cohort FE		✓	✓	✓	✓
Occupation FE				✓	✓
Industry FE			✓		✓
Census Division FE Available	No	Yes	Yes	Yes	Yes
Year FE Available	No	Yes	Yes	Yes	Yes
Race FE Available	No	Yes	Yes	Yes	Yes
Cohort FE Available	No	Yes	Yes	Yes	Yes
Occupation FE Available	No	No	No	Yes	Yes
Industry FE Available	No	No	Yes	No	Yes
R-squared	0.016	0.020	0.031	0.034	0.037
N	99542	99542	99542	99542	99542

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 18: Alpha Regressions: Stepwise Selection Results (Coefficients x 100), Annual Earnings (fearn)

	(1)	(2)	(3)	(4)	(5)
	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)
Age 30-37	-0.0169*** (0.00361)				
Age 62-69	0.0492*** (0.0111)	0.0359** (0.0114)	0.0314** (0.0114)	0.0326** (0.0114)	0.0241* (0.0116)
Bachelor's Degree	2.365*** (0.421)	2.297*** (0.428)	2.725*** (0.440)	1.309** (0.462)	1.603*** (0.466)
Bachelors+	3.846*** (0.582)	3.574*** (0.594)	4.300*** (0.623)	2.341*** (0.668)	2.633*** (0.673)
Probability of Recession	0.0679*** (0.00882)	0.0916*** (0.0199)	0.0872*** (0.0198)	0.0887*** (0.0198)	0.0867*** (0.0198)
5-year moving average of AEP	0.377*** (0.00738)	0.392*** (0.00779)	0.342*** (0.00832)	0.367*** (0.00848)	0.365*** (0.00879)
Out of Labor Force	27.91*** (1.436)	27.42*** (1.435)	20.78*** (1.445)	21.26*** (1.443)	20.41*** (1.444)
Tenure	-0.195*** (0.0228)	-0.116*** (0.0241)	-0.0707** (0.0244)	-0.0710** (0.0244)	-0.0610* (0.0248)
Age 22-29	-0.0289*** (0.00428)		-0.0111* (0.00467)		-0.0101* (0.00470)
Age 38-45				0.00891* (0.00367)	
Age 54-61					-0.0115* (0.00584)
Constant	-0.0140** (0.00433)	-0.0143 (0.0206)	-0.0459* (0.0213)	-0.0483* (0.0244)	-0.0429 (0.0244)
Census Division FE		✓	✓	✓	✓
Year FE		✓	✓	✓	✓
Race FE		✓	✓	✓	✓
Cohort FE		✓	✓	✓	✓
Occupation FE				✓	✓
Industry FE			✓		✓
Census Division FE Available	No	Yes	Yes	Yes	Yes
Year FE Available	No	Yes	Yes	Yes	Yes
Race FE Available	No	Yes	Yes	Yes	Yes
Cohort FE Available	No	Yes	Yes	Yes	Yes
Occupation FE Available	No	No	No	Yes	Yes
Industry FE Available	No	26	Yes	No	Yes
R-squared	0.038	0.041	0.054	0.055	0.057
N	97886	97886	97886	97886	97886

Standard errors in parentheses

Table 19: Alpha Regressions: Lasso Selection Results, Hourly Earnings (fhwage)

	No Controls	No Occ/Ind	No Occ	No Ind	All Controls
EDU1	12	6	10	10	8
EDU2	9	9	8	7	6
EDU3	8	7	6	11	9
EDU4	6	5	5	6	5
OLF	2	3	3	3	3
PrRecess	11				
agecat1	5	8	7	5	5
agecat2	7	11	11	12	10
agecat3	13	10	9	9	7
agecat5	10	11	13	13	11
agecat6	3	4	4	4	4
ma5aep	1	2	2	2	2
tenure	4	12	12	8	9
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes

Table 20: Alpha Regressions: Lasso Selection Results, Annual Earnings (fearn)

	No Controls	No Occ/Ind	No Occ	No Ind	All Controls
EDU1	10	11	13	10	10
EDU2	10	10	11	12	12
EDU3	8	6	5	7	6
EDU4	6	4	4	5	5
OLF	2	3	3	3	3
PrRecess	4				
agecat1	7	8	7	7	7
agecat2	9	9	12	11	11
agecat3	12	7	8	6	6
agecat5	11	12	10	9	9
agecat6	5	5	6	4	4
ma5aep	1	2	2	2	2
tenure	3	5	9	8	8
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes

6 Conclusion

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A Appendix